

with many Internet services such as email, Dropbox and Twitter. It also makes your sensors work together, so they and the rules they activate, are smarter.

The WigWag sensor block is a universal sensor device with eight environmental sensors, four control features, and two expansion ports. It can measure light; detect motion, sound, temperature, humidity, movement, and contact closure (to set up trip-wires). You can also use the extension ports to add more devices, like garden monitors, for instance. The WigWag Glowline is a long LED strip with a power supply that may be used for ambient lighting or for detecting light and motion.



To pair a WigWag device with the app, you just need to scan its QR code with your phone. WigWag also works with a lot of other devices from other manufacturers like Phillips and Belkin. It is also compatible with Raspberry Pi and Arduino, so geeks can do a lot with it! There are different packages available—which include some or all of the devices, as you may want to skip one of these and make use of something you already have, or mix-and-match with other devices.

The Open Twist: Since much of WigWag's hardware and software are open source, developers can go deep into it. WigWag hardware devices are based on Contiki, the open source operating system for the IoT. The rules built with the WigWag app actually generate Javascript; by tweaking this, developers can make the system follow many more rules than the pre-programmed ones. Most of the platform is also extensible via Javascript, thanks to DeviceJS, an open source runtime for executing Javascript built by the WigWag team. It is built on Google V8 and Node.js. DeviceJS ties Javascript to the physical world, by providing hooks for protocols like IP, LoWPAN, Bluetooth, Zigbee, RS-232, and devices such as the Belkin WeMo and Phillips Hue light bulbs. Using DeviceJS, developers can easily make WigWag talk to any device.

The developer kits also include Arduino and Raspberry Pi shields, so you can use WigWag to add Internet-connectivity to a wide range of hardware projects.

Around the shipping date in November 2013, software for the Arduino and Raspberry Pi development boards will be available on Github, and hardware designs for the development kits will be provided.

Converting e-waste into educational toys

<http://dhairyadand.com/thinkertoy/>

MIT Media Labs researcher Dhairya Dand was pained by the amount of e-waste he saw at a landfill in Phnom Penh. He was even more upset by the living conditions of the families that migrated from villages to work at these landfills, heavy with the stench of lead and asbestos. And amongst them were children, even younger than 10 years, who did not find school fun! As an engineer and designer, Dand set about thinking of how he could convert the keyboards and mice thrown away at such landfills into interesting learning tools that would motivate the children of families working there, to learn. He launched ThinkerToys, a project that aims to use money from CSR funds to produce sub US\$ 5 kits that can be distributed to the kids at landfills. These kits simply need to be plugged into keyboards, speakers or mice found at the landfills, to make interesting learning tools.

The Open Twist: Dand has currently designed four toys. Keyano converts old keyboards into pianos; Randomath is a small kit with an LED screen, which transforms a keyboard into a math puzzle generator that kids can use to learn addition and subtraction; Storynory, when plugged into a speaker or headphones, delivers beautiful stories in local languages; and TV++ is a power-packed minicomputer that connects to any old television. All four prototypes are made using Arduino, with off-the-shelf standard components, and can be put together at a very low cost. They are open source, and the community is welcome to improve the designs or even develop new toys. The final products will also remain open source, made from SMD components.



Dand demonstrates his work at expos and exhibitions hoping to garner more support from engineers, entrepreneurs, etc, to make his dream a reality. You can visit the OpenToys community at <https://groups.google.com/forum/?hl=en#!forum/opentoy>